Problem Statement No. 9 – AI Agent for Smart Farming Advice

The Challenge

Agriculture is the backbone of many economies, yet farmers often struggle to access accurate, timely, and reliable information for making informed farming decisions. Guidance related to crop diseases, fertilizer selection, soil health, irrigation methods, weather conditions, and sustainable farming practices is scattered across multiple sources, making it difficult for farmers to obtain trustworthy recommendations quickly.

Many farmers, especially those in rural areas, lack easy access to agricultural experts. As a result, incorrect decisions regarding crop management, fertilizer usage, and disease treatment can reduce crop yield, increase production costs, and negatively impact farm productivity.

To address these challenges, there is a need for an intelligent AI-powered assistant that can provide personalized, accurate, and farmer-friendly agricultural guidance in multiple languages using trusted knowledge sources and modern AI technologies.

The Objective

Develop an **AI-powered Smart Farming AI Assistant** using **IBM watsonx Orchestrate** and **IBM Granite** to provide real-time agricultural guidance for farmers.

The solution should:

- Provide intelligent answers related to crop diseases, fertilizers, soil health, irrigation, weather, and crop management.
- Generate accurate responses using IBM Granite through IBM watsonx Orchestrate.
- Support multilingual conversations in **English, Hindi, and Marathi**.
- Deliver simple and farmer-friendly recommendations for better decision-making.
- "Provide easy access through a responsive React-based web application integrated with IBM watsonx Orchestrate.
- Promote sustainable farming practices and improve agricultural productivity.

Technology

- IBM watsonx Orchestrate
- IBM Granite Model
- IBM Cloud Lite
- Retrieval-Augmented Generation (RAG)
- React.js
- Vite
- JavaScript
- HTML
- CSS
- GitHub
- Vercel

Project Name: Smart Farming AI Assistant

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